

Student Performance Analytics

Business Intelligence Project

Open University Learning Analytics Dataset (OULAD)

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Github link:

<https://github.com/LearningProgram-MIS/BI-Open-Learning-Analytics>

Business Problem & Key Questions

Business Problem:

- Universities face high student withdrawal rates
- Raw learning data is complex and unstructured
- Decision-makers lack clear performance indicators

Key Questions:

- Which students are most at risk of withdrawal?
- Which regions and demographic groups show lower completion and pass rates?
- How can BI help identify performance issues early to support intervention?

Student Performance Analytics

Technical Presentation

Total students	Total Enrollments	Average Score	Completion Rate...	Withdrawal Rate...
29K	33K	75,72	47,20	31,16

Open University Learning Analytics Dataset (OULAD)

28,785 Unique Students	173,912 Assessments	Oct 2012 – Dec 2014 Time Range
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Project Overview

PROJECT background

Online learning platforms generate large volumes of data related to students, courses, and assessments.

However, raw educational data is difficult to analyze without proper transformation. Business Intelligence helps convert this raw data into meaningful insights that support academic decision-making and improve student outcomes.

PROJECT SCOPE

Objective

The main objectives of this project are:

- Analyze student enrollment and academic performance
- Measure completion, pass, and withdrawal rates
- Study the impact of demographics on student success
- Build a BI-ready data model for interactive dashboards

Tools Used

Python (jupyter notebook) • Power BI • DAX

Deliverables

ETL Pipeline • Data Model • Dashboard • Analysis Report

Time Range: Oct 2012 – Dec 2014

Project Overview

1. Data Source & Business Context

2. ETL Pipeline Architecture

3. Star Schema Data Model

4. KPIs & DAX Measures

5. Power BI Dashboard

6. Key Findings & Recommendations

Data Source: OULAD Dataset

Our data exploration is based on a public open learning dataset sourced from Kaggle..
The Open University Learning Analytics Dataset contains anonymized data about courses, students, and their interactions with the Virtual Learning Environment (VLE) for courses presented between 2012-2014.

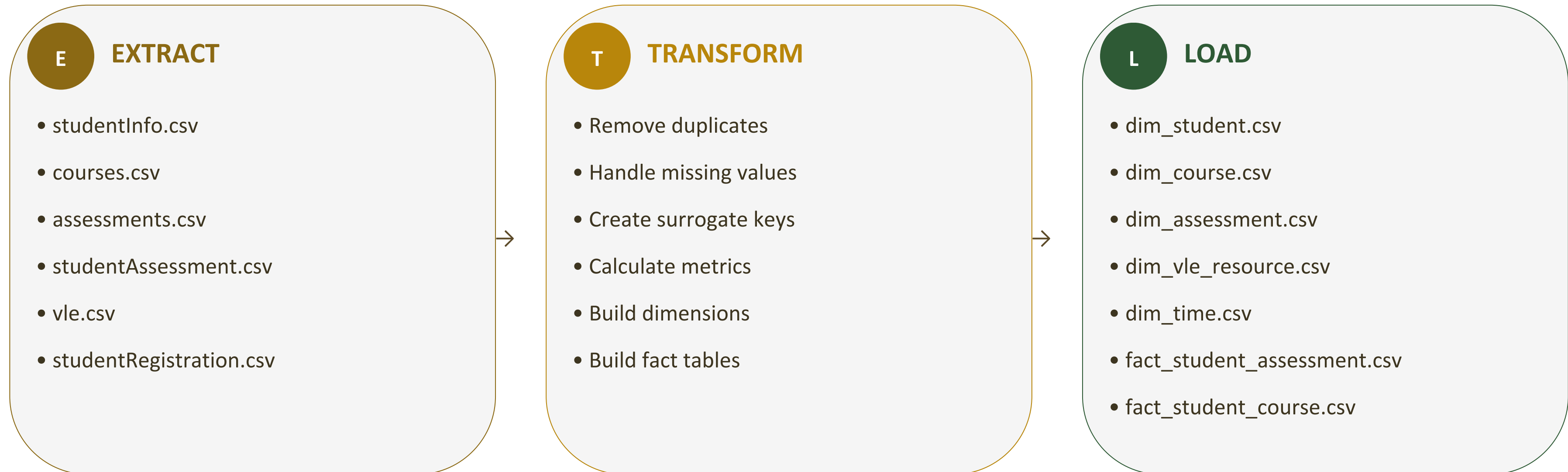
Source Files Extracted

StudentInfo.csv 32,593 Student demographics	Courses.csv 22 Course presentations	Assessments.csv 206 Assessment types
StudentAssessment.csv 173,912 Student scores	Vle.csv 6,364 VLE resources	StudentRegistration.csv 32,593 Registrations

Key Student Attributes: Gender, Age Band, Region, IMD Band, Highest Education, Disability

Final Result Categories: Distinction, Pass, Fail, Withdrawn

ETL Pipeline Architecture



In the load phase, the transformed dimension and fact tables are persisted as CSV files, making them ready for downstream consumption by business intelligence tools such as Power BI and Tableau.

✓ *Data Quality & Validation*

1

Missing values handled

2

Duplicates removed

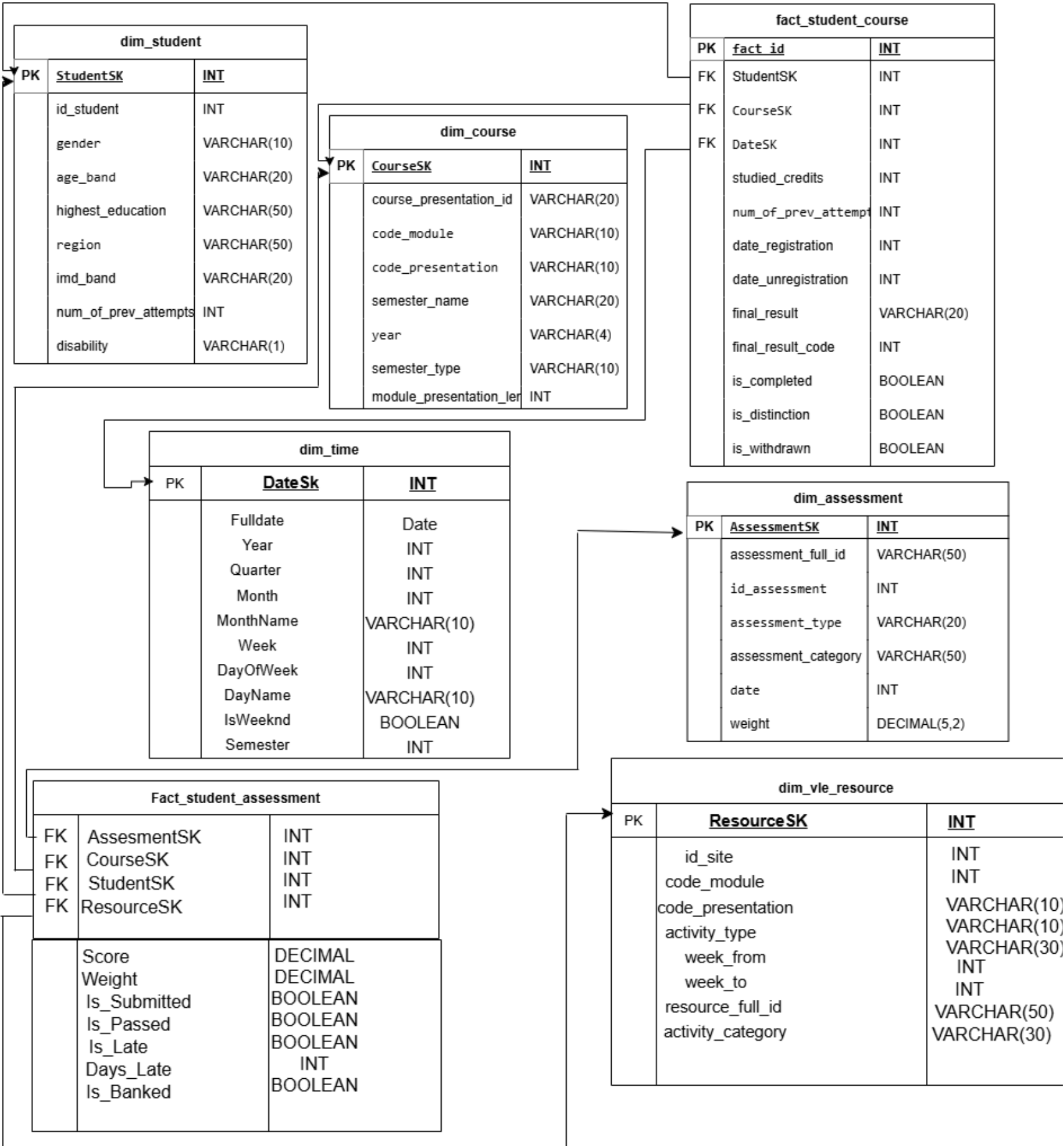
3

Data validation rules applied

4

Score ranges validated (0–100)

Star Schema Data Model



Fact Table Measures

fact_student_assessment

score, weight, is_passed, is_submitted, is_late, Days_late, is_banked

fact_student_course

studied_credits, num_of_prev_attempts, date registration, date_unregistration, final_result, is_completed, is_withdrawn, is_completed, is_distinction

Dimension Table Attributes

dim_student

StudentSK, gender, age_band, region, imd_band, highest_education

dim_course

CourseSK, code_module, presentation_length

dim_assessment

AssessmentSK, assessment_type, date, weight

dim_time

date, year, quarter, month, week, day, semester

dim_vle_resource

activity type, category, weeks, module

KPIs & DAX Measures — Core Metrics

Student & Enrollment

Total Students

```
DISTINCTCOUNT(dim_student[StudentSK])
```

Total Enrollments

```
COUNTROWS(fact_student_course)
```

Active Enrollments

```
COUNTROWS(FILTER(..., is_withdrawn=0))
```

Enrollements Per Students

```
[Total Enrollments] / [Total Students]
```

Performance Metrics

Average Score

```
AVERAGE(fact_student_assessment[score])
```

Pass Rate (%)

```
SUM(is_passed) / COUNTROWS(...) × 100
```

Fail Rate (%)

```
100 - [Pass Rate]
```

Weighted Avg Score

```
SUMX(score × weight) / SUM(weight)
```

Progress & Outcome

Completion Rate (%)

```
SUM(is_completed) / COUNTROWS(...) × 100
```

Withdrawal Rate (%)

```
SUM(is_withdrawn) / COUNTROWS(...) × 100
```

Retention Rate (%)

```
100 - [Withdrawal Rate]
```

Assessment Metrics

Submission Rate (%)

```
SUM(is_submitted) / COUNTROWS(...) × 100
```

On-Time Rate (%)

```
On-time / Submitted × 100
```

Average Days Late

```
AVERAGE(days_late WHERE is_late=1)
```

75.72

Avg Score

47.20%

Completion

31.16%

Withdrawal

9.3%

Distinction

68.84%

Retention

KPIs & DAX Measures — Composite & Demographics

Composite KPIs

Student Performance Index

Weighted composite combining academic performance with submission behavior

$$\begin{aligned} &= 0.5 \times \text{Score} \\ &+ 0.3 \times \text{Submission Rate} \\ &+ 0.2 \times \text{On-Time Rate} \end{aligned}$$

Module Success Score

Overall health indicator for course modules based on student outcomes

$$\begin{aligned} &= (\text{Pass Rate} \\ &+ \text{Retention Rate} \\ &+ \text{Completion Rate}) / 3 \end{aligned}$$

Demographic Metrics

Students by Gender (%)

$$\text{Students per gender} / \text{Total} \times 100$$

Students by Region (%)

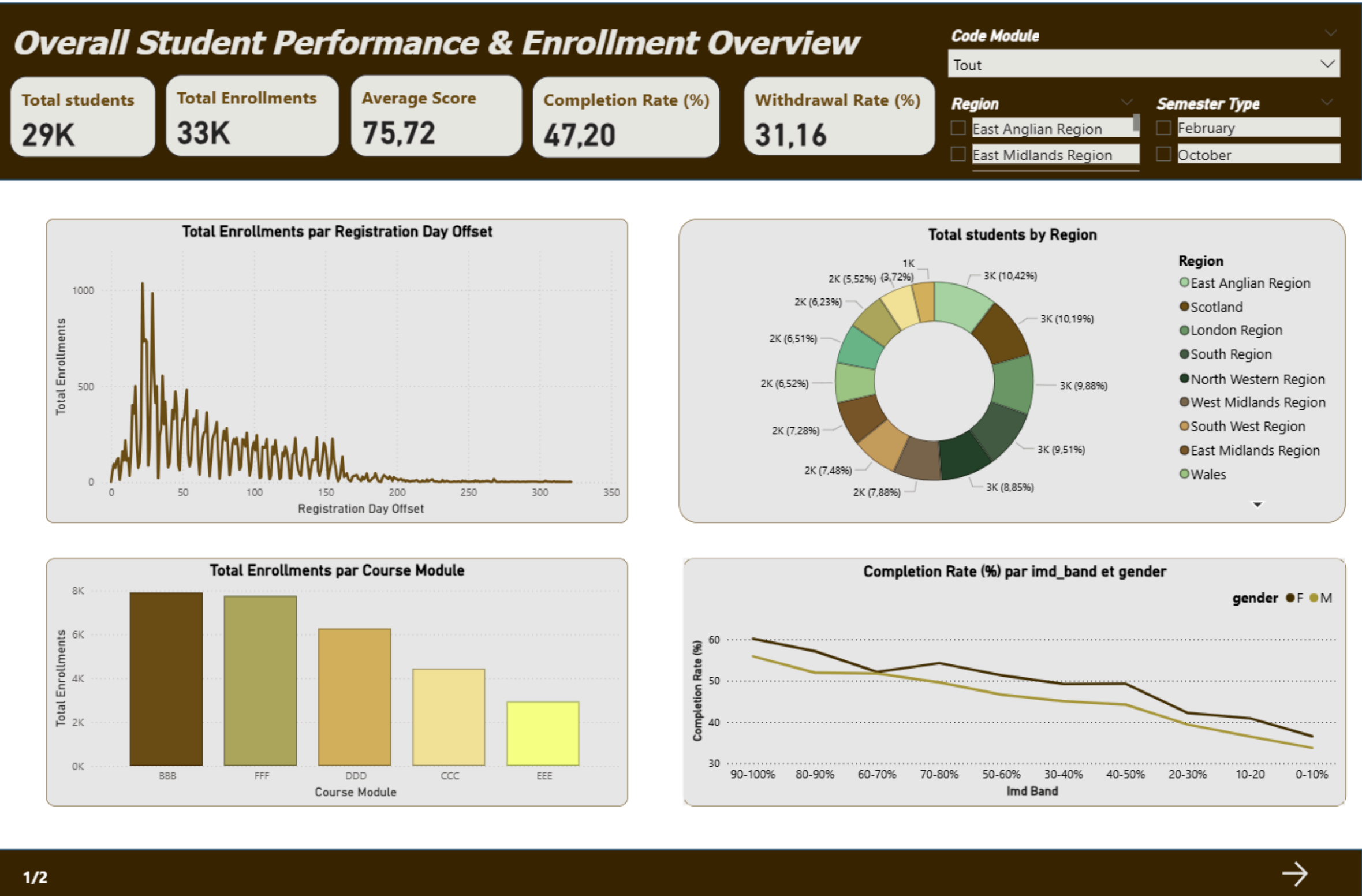
$$\text{Students per region} / \text{Total} \times 100$$

Completion by IMD Band

Completion rate grouped by IMD band

What is IMD Band? Index of Multiple Deprivation measures socioeconomic deprivation in England. Bands range from 0-10% (most deprived) to 90-100% (least deprived). This socioeconomic indicator correlates strongly with student completion rates.

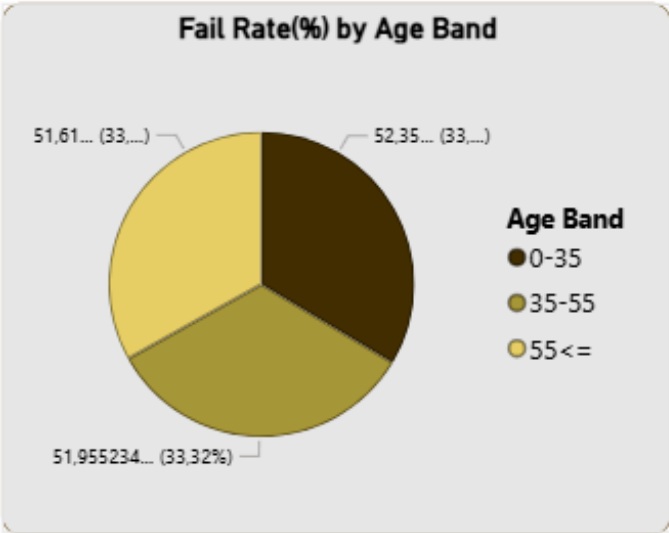
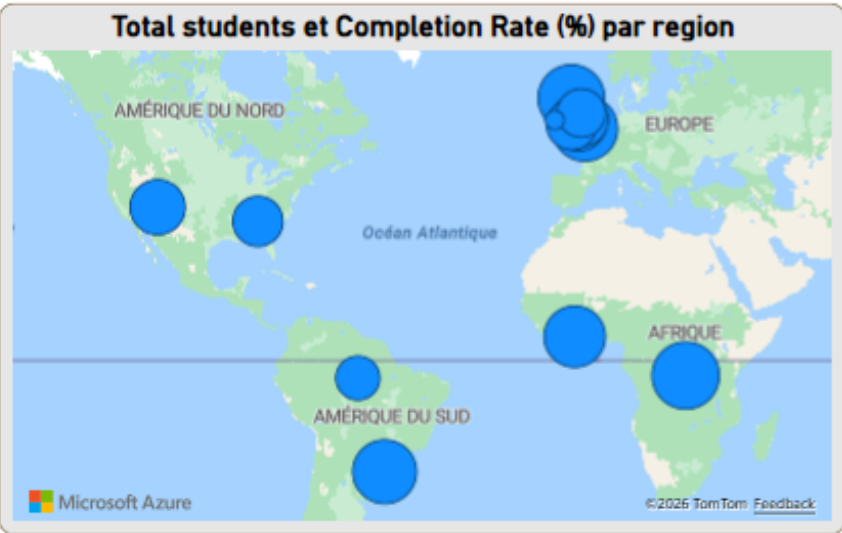
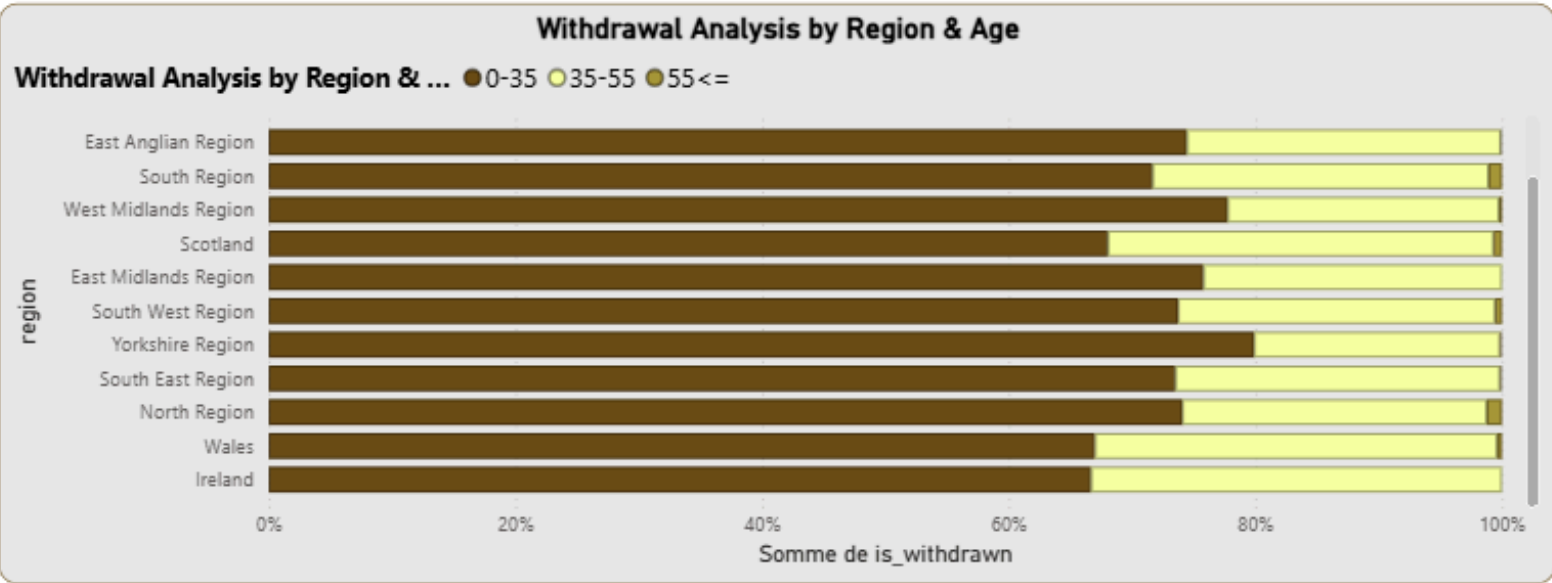
Power BI Dashboard Design



Power BI Dashboard Design

Risk Identification & Student Performance Drivers

age_band	A Level or Equivalent	HE Qualification	Lower Than A Level	No Formal quals	Post Graduate Qualification	Total
0-35	50,61	54,75	35,96	30,61	66,67	44,95
35-55	56,46	57,60	45,80	24,53	62,94	52,34
55<=	57,50	62,88	50,00		100,00	61,75
Total	52,03	56,17	38,85	29,68	65,50	47,20



Dashboard Components

table: Withdrawal Rate by Age Band & Qualification

Stacked Bar: Withdrawal by Region & Age

Map: Total Students & Completion Rate (%) by Region

Pie Chart: Fail Rate (%) by Age Band

Filters: Code Module, Region, Semester Type

This page answers the question: Why are students failing or withdrawing, and who is most at risk?

Key Findings from Analysis

1

Critical Retention Crisis

Despite average score of 75.72, only 47.20% completion with 31.16% withdrawal indicates non-academic barriers

2

Geographic Disparities

London, North Western, and West Midlands show highest withdrawal concentrations

3

Youth Demographic at Risk

The 0-35 age band dominates withdrawal statistics across nearly all regions

4

25-Point Socioeconomic Gap

students in **90-100% IMD** band achieves ~60% completion vs ~35% for **0-10% IMD** band

Gender Gap Secondary

Female 3-5% higher completion, but smaller than socioeconomic effects

Module Imbalance

BBB (~8,000) and FFF (~7,500) dominate enrollments, create resource allocation challenges

Qualification Performance

HE Qualifications shows higher fail rates (56.17%) than A- Level (52.03%)

Strategic Recommendations

- 1

First-Year Experience Program

Target 0-35 age with mentoring. 15% withdrawal reduction projected.
- 2

Regional Support Hubs

Create dedicated centers in London, North West, West Midlands with tutoring.
- 3

Financial Support for Low IMD

Emergency grants and scholarships for 0-40% IMD bands.
- 4

Predictive Analytics System

Early warning system analyzing attendance, assignment submission and assesement performance to flag at-risk students by week 4.

- 5

Redesign High-Enrollment Modules

Audit BBB and FFF modules. Expand sections, enhance support.
- 6

Male Student Initiative

Male-specific mentoring to address 3-5% completion gap.
- 7

Rebalance Module Capacity

Redistribute faculty based on enrollment demand.
- 8

Best Practice Exchange

Transfer practices from high-performing regions quarterly.

Implementation Timeline

PHASE 1 • Months 1-3
Early warning, first-year program

PHASE 2 • Months 4-6
Regional hubs, financial support

PHASE 3 • Months 7-12
Module redesign, rebalancing

Conclusion & Expected Outcomes

This analysis reveals a three-dimensional crisis: retention failures despite adequate academic performance, severe geographic disparities, and significant socioeconomic inequality.

Strategic Imperatives

Invest in First-Year	Eliminate Financial Barriers	Build Regional Hubs	Deploy Analytics
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Expected Outcomes

30-40%	10-15%	£3-4M	£10M+
Withdrawal Reduction	Completion Increase	Investment Required	Revenue Saved

Thank You !

Questions & Discussion

29K

Students Analyzed

7

Tables Created

12+

KPIs Defined

8

Recommendations

Python

Pandas

Power BI

DAX